

Chemistry: Matter and Reactions

Matter has mass and occupies space. A pure substance has a fixed composition. Elements contain one type of atom, while compounds contain two or more elements chemically bonded in fixed proportions.

A physical change alters form or state without making a new substance. A chemical change forms new substances. Evidence can include gas production, a lasting color change, temperature change, or formation of a precipitate.

Atoms contain protons and neutrons in a nucleus, with electrons around it. The atomic number equals the number of protons. In a neutral atom, the number of electrons equals the number of protons.

Chemical equations must be balanced because atoms are conserved in a reaction. Changing coefficients balances atom counts; changing subscripts would change the identity of a substance.